CalcsLive Agent

Design and review CalcsLive calculations, create persistent articles, and bridge them to/from Excel when connected.

Bridge to/from Excel

Send Calc to Excel Get Calc from Excel

Two-way bridge: send a reviewed CalcsLive calc to Excel, or read + convert + review an Excel table as a CalcsLive script.

How to use this bridge

- **Send Calc to Excel**: First review and create a calculation article in the app. Then send that created article into Excel so the bridge writes the metadata block and PQ table structure for interactive use.
- **Get Calc from Excel**: Prepare a compatible PQ table in Excel, then let the bridge read it, convert it into a stateless CalcsLive script review payload, and show the reviewed result back in the app before you persist it as a new article.
- **Excel table structure expected**: the bridge looks for columns matching Description, Symbol, Expression, Value, and Unit. A left-side row number column is allowed. For inputs, leave Expression blank and provide initial Value + Unit. For outputs, provide an Expression and desired Unit.
- **Recommended flow for Excel-originated calculations**: lay out the PQ rows in Excel using the same table structure, click **Get Calc from Excel**, review warnings/results, adjust Article Title and Description if needed, then click **Create Article**.

Reviewed CalcsLive Script

Summary: Calculation completed with 2 outputs

Mass of Solid Cylinder Calculator

	Kind	Description	Symbol	Expression	Value	Unit
0	Input	Density of the material	rho		7,850.0000	kg/m^3
1	Input	Radius of the cylinder	D		0.5000	m
2	Input	Height of the cylinder	h		12.0000	in
3	Input	Thickness	t		10.0000	mm
4	Output	Mass of the cylinder	m1	PI*D^2/4*h*rho	469.8016	kg
5	Output	Mass of topless cylinder	m2	PI*(D^2*h-(D-*t*2)^2*(h-t))/4*rho	51.0375	kg

Category Metadata

Create Calculation Article

Create Article

The last reviewed `run_calcslive_script` payload is ready for approval.

Article Title

Mass of Solid Cylinder Calculator

Article Description

Calculates the mass of a solid cylinder based on density, radius, and height.


```
{
  "pqs": [
    {
      "sym": "rho",
      "unit": "kg/m\u00b3",
      "description": "Density of the material",
      "value": 7850.0
    },
    {
      "sym": "D",
      "unit": "m",
      "description": "Radius of the cylinder",
      "value": 0.5
    },
    {
      "sym": "h",
      "unit": "in",
      "description": "Height of the cylinder",
      "value": 12.0
    },
    {
      "sym": "m1",
      "unit": "kg",
      "description": "Mass of the cylinder",
      "expression": "PI*D^2/4*h*rho"
    },
    {
      "sym": "t",
      "unit": "mm",
      "description": "Thickness",
      "value": 10.0
    },
    {
      "sym": "m2",
      "unit": "kg",
      "description": "Mass of topless cylinder",
      "expression": "PI*(D^2*h-(D-t*2)^2*(h-t))/4*rho"
    }
  ],
  "inputs": {
    "rho": {
      "value": 7850.0,
      "unit": "kg/m\u00b3"
    },
    "D": {
      "value": 0.5,
      "unit": "m"
    },
    "h": {
      "value": 12.0,
      "unit": "in"
    },
    "t": {
      "value": 10.0,
      "unit": "mm"
    }
  },
  "outputs": {
    "m1": {
      "unit": "kg"
```

```
    },
    "m2": {
      "unit": "kg"
    }
  },
  "title": "Mass of Solid Cylinder Calculator",
  "description": "Calculates the mass of a solid cylinder based on density, radius, and height."
}
```


Last Created Article

Title: Mass of Solid Cylinder Calculator

ID: `3MMT7FNSY-2MX`

URL: https://www.calcslive.com/view/3MMT7FNSY-2MX

Modes: edit | calculate | table | view

Summary: Calculation completed with 2 outputs

	Kind	Description	Symbol	Expression	Value	Unit
0	Input	Density of the material	rho		7,850.0000	kg/m³
1	Input	Radius of the cylinder	D		0.5000	m
2	Input	Height of the cylinder	h		12.0000	in
3	Input	Thickness	t		10.0000	mm
4	Output	Mass of the cylinder	m1	PI*D^2/4*h*rho	469.8016	kg
5	Output	Mass of topless cylinder	m2	PI*(D^2-(D-2*t)^2*(h-t))/4*rho	51.0375	kg

Raw response

Hi! I can help you design a CalcsLive calculation, review it, persist it, and, when Excel is connected, bridge it to and from Excel with automatic updates.

Calculate the mass of a solid cylinder.

The calculation for the mass of a solid cylinder ran successfully with the default inputs I provided. Here's a summary of the results:

Calculation Summary

- **Inputs Used:**
  - Density (ρ): 1000 kg/m³ (e.g., typical for water)
  - Radius (r): 0.5 m
  - Height (h): 1.0 m
- **Output:**
  - Mass (m): 785.398 kg

This was based on the formula: mass = density × π × radius² × height.

Details from the Run

- **Execution Time:** 23 ms
- **Warnings:** None – all units were recognized and consistent (e.g., density in DENSITY category, radius and height in LENGTH category, mass in MASS category).
- **Human-Readable Output:** The mass is approximately 785.4 kg with the given inputs.

If you'd like to:

- **Change the inputs or units** (e.g., use a different density or imperial units like pounds and inches), just let me know, and I can update and re-run the calculation.
- **Persist this calculation** as a reusable CalcsLive article, I can help review and create it – but I'll only do so if you explicitly confirm.
- **Load this into Excel** for editing and live updates (if Excel is connected), I can check the connection and set it up.

What would you like to do next?

Ask me to 'Calculate the values and write them back to Excel'...